

Machine Learning–Guided Screening of Magnetic Materials for Thermally Robust Non-Volatile Data Storage

Project status report

Prepared for advisor review · 15 August 2026

1. Project goal

The project screens magnetic materials — computationally and, where candidates warrant it, via spin-dynamics simulation — for systems that host stable, controllable spin textures (e.g. spin spirals, skyrmion-hosting interfaces such as Pd/Fe/Ir(111)-type stacks) suitable for thermally robust non-volatile memory. The approach combines (a) a large harmonized materials database, (b) an ML classifier for magnetic ordering that corrects for known sampling bias in existing databases, (c) a machine-learned interatomic potential (MLIP) fine-tuned for magnetic systems, and (d) atomistic spin-dynamics simulation of shortlisted candidates.

The pipeline has five phases:

Phase	Purpose	Status
0 — Data harmonization	Merge multiple public materials databases into one schema	7 sources merged, 64,635 records (§2)
1 — Bias-corrected ordering classifier	Predict magnetic ordering (FM/AFM/FiM/NM) from composition + structure, corrected against MAGNDATA ground truth	Structure-aware v2 trained, 67.0% accuracy (§3)
2 — Candidate screening	Apply the classifier to the whole database, filter for DMI-compatible candidates, backfill structures, relax with MACE	298 candidates relaxed, final shortlist produced (§4)
3 — MLIP fine-tuning	Fine-tune a MACE foundation model for fast, accurate energetics on magnetic systems	Training run complete (60/60 epochs)
4 — Spin dynamics & DFT validation	LAMMPS-SPIN simulation of shortlisted candidates (exchange + DMI + precession), and real VASP DFT magnetic-ordering energy mapping	Exploratory spin dynamics on one system; DFT validation campaign in progress (§6.3)

2. Phase 0 — Data harmonization

All sources are merged into one harmonized schema (`material_id` , `formula` , `elements` , `space_group` , `density` , `volume` , `band_gap` , `ordering` , `total_magnetization` , and — where available — full `structure` and per-site `magmom` vectors, the last two being required inputs for Phases 3 and 4). Records keep their source database tagged rather than being silently deduplicated; conflict resolution across sources is a deliberate downstream step, not baked into acquisition.

Source	Records	Fields	Notes
Materials Project	37,783	full schema	via <code>mp-api</code>
JARVIS-DFT	19,116	full schema	
AFLOW	2,639	full schema	
MAGNDATA (rescraped)	1,919 (of 2,414 attempted)	full schema + structure + per-site <code>magmom</code>	See §2.1
2DMatPedia	1,862	full schema + structure	

			the project's only 2D-materials source so far; closes the "2D screening" gap for this source specifically (C2DB remains unacquired)
OQMD	816	composition + stability (no magnetism fields — not exposed by this endpoint)	cross-validation role only, per project proposal
NOMAD	500	full schema	
Total (current)	64,635		3,781 with full structure (up from 1,919 pre-merge)

Not yet acquired: C2DB. None of OQMD/NOMAD/2DMatPedia carry a magnetic `ordering` label — merged in as unlabeled (`ordering=None`) per the same anti-fabrication principle already applied to JARVIS (§3.1): never hand-derive a label from a net-moment proxy, leave it for the classifier to predict.

Bug found and fixed: OQMD's `elements` and `space_group_number` fields were silently hardcoded to `None` in the acquisition script despite the raw source fields (`composition` , `spacegroup`) being present and parseable — discovered because the screening funnel (§4.1) was dropping exactly 816 records, OQMD's full count. Fixed by deriving both fields properly (via `pymatgen.core.Composition` and `pymatgen.symmetry.groups.SpaceGroup` , the latter needing an underscore-notation fallback for screw-axis symbols like "P21/m" → "P2_1/m"); 816/816 records now have populated `elements` , 815/816 have `space_group_number` (one symbol still unresolved by either notation, low priority). This unblocked 32 OQMD candidates in the screening funnel (§4.1).

2.1 MAGNDATA rescrape (complete)

The only source in the dataset with **ground-truth** magnetic structure (not a composition-based guess) is MAGNDATA (Bilbao Crystallographic Server), and the existing scrape covered only 79–102 records with no surviving acquisition script or documentation of method. This is a hard blocker for both the Phase 1 classifier (which needs real labels to correct its composition-only bias) and Phases 3–4 (which need per-site magnetic moments as simulation input). A new scraper, `acquire_magndata.py` , was written and empirically verified end-to-end against the live server rather than assumed:

- Full index endpoint returns all entries in one page (no pagination) — **2,414 unique entries** found, ~24× the previous scrape, spanning four ID-family prefixes (`0.x` : 1,155, `1.x` : 883, `2.x` : 118, `3.x` : 30).
- Structure files (MCIF format) are directly downloadable without visiting a per-entry detail page first, once the exact filename-construction rule was reverse-engineered and verified against edge cases (formula strings with HTML subscript markup).
- Per-site magnetic moment vectors, space group (BNS number), and full crystal structure are parsed via `pymatgen` ; magnetic ordering (FM/AFM/FiM/NM) is derived with a heuristic on the net vs. summed moment magnitude, since MAGNDATA does not label ordering directly. Validated against a known case (LaMnO₃, correctly recovered as AFM with space group 62/Pnma).

Data quality finding: of the 495 permanent failures (2,414 attempted, 1,919 succeeded, 495 failed — 20.5% failure rate), the large majority (466) are disordered/partial-occupancy magnetic structures that `pymatgen`'s CIF parser cannot currently represent (`NotImplementedError`) — a real upstream limitation, not a scraper bug. The remainder are a small number of malformed source records (e.g. an unparseable numeric field, `ValueError` / `AttributeError`) — worth flagging for anyone downstream expecting full MAGNDATA coverage, but not something further retries can fix.

Final result: 1,919 / 2,414 entries successfully parsed and merged into the harmonized dataset (all 2,414 index entries resolved — none pending). One resume-logic bug was found and fixed along the way (a prefixed-vs-bare ID mismatch was causing permanent failures to be needlessly re-fetched and re-logged on every resume, inflating the failure log without affecting the successfully-collected records); after the fix, a handful of previously-flaky URLs (8 of 21) resolved successfully on retry, confirming most tmp/-path 404s really are transient server-side flakiness rather than missing files.

3. Phase 1 — Magnetic ordering classifier

A LightGBM baseline predicts ordering (FM/AFM/FiM/NM) from composition-derived features only (no structure input yet). Retrained after the MAGNDATA rescrape landed (§2.1), which grew the MAGNDATA-labeled portion of the training set 24× (79 → 1,919 records).

Metric	Before rescrape	After rescrape (current)
Accuracy	56.7%	56.7%
Balanced accuracy	61.8%	60.7%

Train / test size	27,234 / 6,809	28,769 / 7,193
Class balance (train+test)	FM 22,137 · FiM 6,035 · NM 3,730 · AFM 2,141	FM 22,259 · FiM 6,238 · AFM 3,735 · NM 3,730

Honest finding: a 24× increase in ground-truth MAGNDATA data left accuracy essentially flat and balanced accuracy slightly *lower*. This is not noise — it is a genuine ceiling, and the cause is identified: the feature set is **composition-only** (per-element statistics: mean/min/max/range/std of atomic number, mass, electronegativity, atomic radius, plus group/period/transition-metal/rare-earth fractions). It never consumes the `structure` or per-site `magmom` fields that the new MAGNDATA records actually carry, so more MAGNDATA rows added more *labels* but no new *signal* the model can use. Top feature importances (`density` , `mean_electronegativity` , `mean_group`) are unchanged in character before and after — further confirming the model is leaning on coarse periodic-table statistics, not real magnetic structure.

3.1 Structure-aware features (v2) — result

Two tiers of structural features were added on top of the v1 composition features. **Tier 1** (broad coverage, ~100% of labeled rows across all sources): space group number, a crystal-system bucket derived from it, unit-cell volume, and site count — fields already present in the harmonized schema but never used by v1. **Tier 2** (narrow, MAGNDATA-only, ~5.3% of labeled rows): local geometry around magnetic-species sites (nearest magnetic-magnetic neighbor distance, coordination number), computed from atomic positions and species only — deliberately *not* from the `magmom` vectors, since MAGNDATA's `ordering` label is itself derived from those vectors and using them as features would leak the label-generating function. Missing Tier 2 values (97% of rows) are left as NaN; LightGBM handles this natively rather than requiring imputation.

Feature set	Accuracy	Balanced accuracy
Composition only (v1)	56.7%	60.7%
+ Tier 1 (symmetry/volume/nsites)	63.9%	65.6%
+ Tier 2 (local structure, MAGNDATA-only)	62.6%	65.0%
+ Tier 1 + Tier 2 (v2, full)	67.0%	68.1%

A real, substantial gain (+10.2 points accuracy, +7.4 points balanced accuracy), and the ablation shows both tiers contribute independently rather than one doing all the work: broad symmetry/volume fields close roughly two-thirds of the gap on their own, but the narrow, MAGNDATA-only local-geometry features add a comparable amount on top despite covering only ~5% of rows — i.e. real atomic-structure information matters, even in this still-limited-coverage form. `space_group_number` is now the single most important feature (previously absent from the model entirely).

New finding, unrelated to this work but material to the whole classifier: JARVIS (19,116 raw records, would be the second-largest source) has **zero** populated `ordering` labels — every JARVIS record's `ordering` field is null, so it contributes nothing to training or evaluation. Of the 61,457 harmonized records, only 35,962 have a usable `ordering` label at all, and of those, Materials Project (34,037) and MAGNDATA (1,919) account for effectively all of it; AFLOW contributes 6. This was true before this session's work and is not a regression from anything changed here — flagged because it means the effective labeled dataset is smaller and less diverse than the raw totals suggest, and because JARVIS-DFT does carry data (e.g. computed magnetic moment) that could plausibly be mapped to an `ordering` label if that's worth pursuing.

4. Phase 2 — Candidate screening, structure backfill, and MACE relaxation

Phases 1, 3, and 4 had each been validated in isolation (classifier trained/evaluated on a held-out split; MACE fine-tuned on the harmonized dataset; spin dynamics run on one hand-picked system) but nothing connected them — there was no step that ran the classifier over the whole database and turned it into a ranked, physically-screened, structurally-verified candidate list. `src/ranking/` was an empty scaffold directory before this work; it now holds the full screen → backfill → relax → finalize pipeline.

4.1 Screening funnel

A deployment classifier (same v2 feature set as §3.1, trained on **all** labeled data rather than the held-out split used for evaluation) predicts `ordering` for every record that lacked a ground-truth label — including all 19,116 previously-unusable JARVIS records (§3.1's JARVIS finding) — then the full 64,635-record database (§2, after the OQMD/NOMAD/2DMPedia merge) is filtered for candidates physically

compatible with a Dzyaloshinskii–Moriya interaction (DMI), the mechanism responsible for stabilizing the chiral spin textures this project targets:

- **Net magnetic moment** — predicted or ground-truth ordering is FM or FiM (AFM excluded; the existing Phase 4 simulation code assumes a net-moment texture).
- **Noncentrosymmetric space group** — DMI is symmetry-forbidden under inversion (Moriya's original criterion). Verified computationally, not from memory: all 230 space groups' symmetry operations were checked for an inversion operator — 92 centrosymmetric / 138 noncentrosymmetric, matching the textbook count.
- **A heavy spin–orbit-coupling element present** (Ta, W, Re, Os, Ir, Pt, Au, Hg, Bi) — the classic DMI-enhancing species in both interfacial (Fert–Levy) and bulk (Moriya) mechanisms. This is a heuristic compositional filter, not a computed DMI magnitude — nothing in the pipeline yet computes an actual DMI value.
- **Thermodynamic plausibility** — `energy_above_hull` ≤ 0.1 eV/atom where known (92.6% coverage, MP + JARVIS only); AFLOW/MAGNDATA lack this field and are kept, ranked below known-stable candidates rather than excluded outright.

Of 64,635 records, 1,647 pass the full funnel (source breakdown: MP 772, AFLOW 386, JARVIS 353, NOMAD 62, 2DMatPedia 40, OQMD 32, MAGNDATA 2).

4.2 Structure backfill

Only MAGNDATA carries full atomic `structure` in the harmonized schema — MP/JARVIS/AFLOW records never had it acquired at the ~60k scale, by design (too costly for records that end up unused). For the top 100 screened candidates per source, structures were fetched via each source's own public data-access method: JARVIS's bulk `dft_3d` figshare dataset (no key), AFLOW's public AFLUX REST API plus a direct `CONTCAR.relx` fetch (no key, endpoint behavior verified live before use), and Materials Project via `mp_api` (requires an API key — none was present in this environment; one was supplied for this run). 299 of 300 attempted backfills succeeded (AFLOW 99/100, JARVIS 100/100, MP 100/100).

4.3 MACE relaxation as a consistency check

Every backfilled structure is already DFT-relaxed at its source database's own DFT settings. Re-relaxing each one with the fine-tuned MACE potential (Stage 2/SWA checkpoint, position-only / fixed-cell, since the model's stress accuracy is not yet validated enough to trust for cell relaxation of novel compositions) and measuring how far it moves is a cheap, independent test of whether the fine-tuned potential agrees the structure is a local minimum on its own learned surface.

Result: 298/298 candidates relaxed without error, 298/298 converged (max force ≤ 0.05 eV/Å). Relaxation-energy shift was near-zero for the overwhelming majority (median 0.0009 eV/atom) — strong agreement between the fine-tuned potential and each source's own DFT. The signal also did real diagnostic work: the worst outliers (relaxation energy up to 1.79 eV/atom) were almost all JARVIS candidates that were *already* flagged as far from the convex hull by DFT (1.1–3.8 eV/atom above hull) — i.e. MACE's independent judgment corroborates the source thermodynamic data rather than contradicting it, which is exactly the behavior wanted from this check.

4.4 Final Phase-4-ready shortlist

The classifier-confidence ranking (§4.1) and the MACE relaxation-consistency signal (§4.3) were merged into one final ranking (converged → hull-stable → smallest MACE relaxation shift → classifier confidence). The top candidate is `mp-1520073` (SrCaFeBiO₆, FM, ground-truth label, cubic, noncentrosymmetric, contains Bi, 0.089 eV/atom above hull, zero relaxation shift). The top 30 are written to `outputs/phase4_shortlist.csv` and are ready for Phase 4 spin-dynamics simulation, pending a priority/selection call on how many to actually simulate.

5. Phase 3 — MACE foundation-model fine-tuning

Fine-tuning MACE's `small` foundation model on the harmonized dataset using multihead training: a `Default` head learns the new (magnetic-materials) data while a `pt_head` replays a 20,000-configuration sample of the original pretraining distribution, to fine-tune on the new domain without catastrophically forgetting the foundation model's general chemistry (standard continual-learning technique for MLIP fine-tuning).

Setting	Value
Foundation model	MACE-small

Precision	float32
Batch size	32
Max epochs	60 (SWA from epoch 45, patience 15)
Loss weighting	energy 1.0 / forces 100.0 / stress 1.0 (Stage 1); energy 1000 / forces 100 / stress 10 (Stage 2, post-SWA)
Replay set	20,000 foundation-distribution configurations (pt_head)

5.1 Performance fix

The initial run defaulted to float64 precision and single-threaded data loading (both silent defaults of `mace_run_train`, not deliberate choices) and was running at ~32 min/epoch — projected ~32h total. Root-caused via the run's own logged configuration, not guesswork. On this GPU (Ampere-class, weaker at FP64 than FP32), switching to float32 with 8 parallel data-loading workers cut this to **~6.3 min/epoch — roughly a 5× speedup** — confirmed to hold steady across all 60 epochs of the completed run, not a one-off. The original float64 run's checkpoints (18 epochs in) were archived rather than discarded.

5.2 Training complete — final results

All 60/60 epochs finished successfully; both a Stage 1 (pre-SWA, epoch 44) and Stage 2 (post-SWA, epoch 59) checkpoint were saved and compiled.

Checkpoint	Split	RMSE Energy	RMSE Forces	RMSE Stress
Stage 1 (epoch 44)	train (Default head)	75.2 meV/atom	104.0 meV/Å	11.6
	valid (Default head)	79.1 meV/atom	111.0 meV/Å	7.8
Stage 2 / SWA (epoch 59)	train (Default head)	41.7 meV/atom	122.6 meV/Å	12.8
	valid (Default head)	51.1 meV/atom	133.4 meV/Å	9.1

The Stage 2 loss reweighting (energy 1000 / forces 100 / stress 10, vs. Stage 1's 1 / 100 / 1) markedly improved energy accuracy (valid RMSE_E 79.1 → 51.1 meV/atom) at a small cost to force/stress accuracy, as intended by that reweighting.

Signal that was being watched, now resolved: during Stage 1, the `pt_head`'s (replay/foundation-distribution) energy RMSE drifted up substantially — from an initial 39.6 up to a peak of 134.4 meV/atom at epoch 43 (a ~3.4× degradation), a genuine partial-forgetting signal, not noise. The Stage 2 loss reweighting largely corrected this: `pt_head` RMSE recovered to 54.2 meV/atom by epoch 58 — close to its starting value. Net assessment: some forgetting occurred during the aggressive Stage 1 fine-tuning phase but was substantially recovered by the Stage 2 reweighting; not a clean zero, but not a runaway failure either.

6. Phase 4 — Spin dynamics and DFT validation

Simulated one candidate interfacial system (a Pd/Fe/Ir(111)-type stack, chosen for its interfacial Dzyaloshinskii–Moriya interaction, the term responsible for stabilizing chiral spin textures like spirals and skyrmions) using LAMMPS-SPIN (exchange + DMI + Zeeman/precession + Langevin thermostat + LBFGS minimization).

6.1 Solver limitation found, root-caused, and fixed

Update: the dynamics-integration segfault reported earlier is now resolved. Rebuilding LAMMPS from source with the SPIN package explicitly enabled (it was silently `OFF` in every pre-existing local build, including newer ones) did *not* fix it — the crash still reproduced on the real system on a brand-new build, disproving the initial "just needs a newer LAMMPS" hypothesis. A systematic one-variable-at-a-time bisection (11 reproducer scripts, smallest first) then isolated the actual trigger: LAMMPS's `pair_style hybrid/overlay` segfaults during force computation whenever it wraps *exactly two* SPIN-package pair substyles — reproduced for exchange+DMI and, independently, for exchange+magelec, and shown to be unrelated to the thermostat, fix declaration order, lattice geometry, or DMI direction. Confirmed as a genuine upstream LAMMPS bug rather than a modeling error: `examples/SPIN/bfo`, an official example shipped with LAMMPS itself that combines *three* spin substyles (exchange+magelec+DMI), runs cleanly on the exact same build. **Workaround:** pad the production Hamiltonian's `hybrid/overlay` with a third, physically-inert (zero-strength) `spin/magelec` substyle — verified against the real 600-atom scaled Pd/Fe/Ir(111) system (500-step Langevin run, smooth non-NaN energy relaxation) before adopting it. Simulated annealing /

finite-temperature dynamics is therefore now available. A coarse 10-point scan (10–300 K) ran first and completed cleanly; a refined 11-temperature $\times$ 3-seed scan (190–245 K, bracketing the reference paper's $T_c \approx 214$ K) followed, using the local 128-core machine via task-level (not MPI) parallelism — MPI domain decomposition was tried first and found structurally incompatible with this system's single-atomic-layer geometry (LAMMPS-SPIN's `fix nve/spin` requires exactly 8 spatial sectors whenever `nprocs>1`, which a quasi-2D slab can never satisfy in the thin dimension, confirmed by reading `src/SPIN/fix_nve_spin.cpp` directly); 33 independent single-core runs via `xargs -P` sidestepped this and also gave real seed-to-seed statistics. All 33 completed with no errors. Energy rises smoothly and monotonically with T (150.4 $\rightarrow$ 156.5 eV, 190 $\rightarrow$ 245 K) as physically expected. The heat-capacity-like signal (dE/dT from the equilibrated half of each trajectory) shows a mild decrease from ~ 0.15 – 0.17 eV/K (195–210 K) to ~ 0.08 – 0.10 eV/K (214–235 K) rather than a sharp peak at 214 K — a real but inconclusive signal; the 2 ps production window is likely too short to resolve critical slowing-down cleanly, and this is reported as such rather than forced into a confirmation of the literature T_c . The multi-seed LBFGS minimization workaround described below remains useful independently (annealing and minimization answer different questions) but is no longer the only option for exploring metastability.

6.2 Findings

- Spin-spiral wavelength is consistent across 5 independent seeds (5.0–5.8 nm, mean ~ 5.4 nm) — rules out a quench artifact; this is a genuine equilibrium property of the currently coded Hamiltonian.
- That wavelength is $\sim 20\%$ longer than the ~ 4.5 nm target reported in the reference literature for this system.
- Propagation direction is not consistently aligned to the crystallographically preferred (Γ –K) directions across seeds — indicative of a multi-domain outcome rather than a single clean spiral.
- Most likely cause of both discrepancies: the DMI term is currently coded as a single uniform bond-independent vector, whereas this class of interfacial system physically requires a bond-direction-dependent (Fert–Levy) DMI form, which LAMMPS's stock `spin/dmi` pair style cannot express via one `pair_coeff`. Flagged, not yet pursued further (parked pending direction on priority).

6.3 Real DFT magnetic-ordering validation (in progress)

Generalizing §6.1–6.2's spin dynamics to new shortlisted candidates (§4.4) requires per-candidate exchange (J) and DMI (D) constants that nothing in this pipeline can honestly produce: the fine-tuned MACE model (§5) was never trained with per-site `magmom` as an input feature (verified directly in `build_finetune_dataset.py` — zero references), so it cannot distinguish FM from AFM orderings of the same atomic structure, and no DFT infrastructure existed in this project until now. Rather than transplant literature J/D values onto an unrelated composition, this uses real compute newly made available on the account's HPC clusters to compute genuine total energies.

Method: the same collinear magnetic-ordering energy-mapping approach behind Materials Project's own magnetic-ordering workflow (Horton, Montoya & Persson, *npj Comput. Mater.* 2019) — enumerate symmetry-distinct FM/AFM spin configurations of a candidate's relaxed structure (`pymatgen`'s `MagneticStructureEnumerator`, backed by `enumlib`), generate Materials-Project-standard VASP inputs for each (`MPRelaxSet` : composition-aware GGA+U, e.g. Fe gets `LDAUU=5.3` eV matching MP's own convention, `ENCUT` 520 eV, Γ -centered k -mesh), and override only `MAGMOM` per ordering. This keeps results directly comparable to the ground-truth MP energies the candidate was screened against, rather than being an arbitrary DFT recipe.

First candidate: `mp-861559` (TaFeO₄, 36 atoms, 6 Fe sites, space group $I4_1md$ #109) — chosen because 6 Fe sites in one cell allow AFM ordering enumeration without building a supercell, keeping the first end-to-end validation pass fast. FM + 3 symmetry-distinct AFM orderings (of 7 the enumerator found) were staged, with per-site signed `MAGMOM` values verified correct against the actual on-disk POSCAR site order (VASP input generation reorders sites by spin-decorated species internally; confirmed the three AFM configurations remain spatially distinct after this reorder, not accidentally collapsed to duplicates).

POTCAR licensing: pseudopotential data is license-restricted and was never copied to this repository or the local machine — only POTCAR *symbol names* (e.g. `Fe_pv`) were generated locally; the actual POTCAR files are concatenated remotely from the account's own licensed pseudopotential library.

Compute: jobs run on `paramsanganak` (IIT Kanpur), VASP 5.4.4, 48 cores/job, submitted under the account's `upamanyuv25` allocation after the primary `faiza21` allocation hit its SLURM submit-limit QOS. One environment issue was found and fixed along the way: the account's established job script convention (`module load compiler/intel/2020.2.254`) no longer works — the cluster's module tree was reorganized and that module name no longer exists, even though the underlying Intel MPI/MKL toolchain `vasp_std` was linked against is still on disk. Verified via `ldd` (5 unresolved libraries: MKL + `libfabric`) that sourcing the Intel-provided `compilervars.sh` / `mpivars.sh` scripts directly resolves all dependencies; job scripts now do this instead of trusting the stale module name.

A second issue surfaced once the initial batch actually started running (past the queue backlog noted above): all 4 jobs crashed mid-run with a 48-rank SIGSEGV inside VASP's own electronic-minimization code (`nonlr_mp_raccmu` $\rightarrow$ `hamiltmu` $\rightarrow$ `eddav` $\rightarrow$ `elmin`), not

a startup or library-path problem. Root cause: the compute nodes' default stack-size limit (8192 KB) is too small for VASP's non-local projector routines at 48-way MPI parallelism, a well-known VASP failure mode. **Fixed** by adding `ulimit -s unlimited` before the `mpirun` line (`src/dft/write_job_scripts.py`); verified against one job (`0_fm`) before resubmitting the other three to avoid burning queue slots on an unconfirmed fix.

Status: all 4 jobs (FM + 3 AFM) completed successfully — every OUTCAR reports "reached required accuracy - stopping structural energy minimisation" with an empty `slurm.err` . Converged total energies:

Ordering	TOTEN (eV)	mag (μ_B)	ΔE vs. FM
FM	-304.60011	30.0	—
AFM-1	-304.87416	0.0	-0.274 eV
AFM-2	-304.78236	0.0	-0.182 eV
AFM-3	-304.74876	0.0	-0.149 eV

Result: the classifier's FM prediction for mp-861559 is not confirmed by DFT. All three AFM orderings tested are lower in energy than FM; the ground state found is AFM-1, 0.274 eV/cell (≈ 45.7 meV/Fe, 6 Fe sites) below FM. This is a genuine, real-compute validation result — the first case in this project where the classifier's ordering call has been checked against ground-truth DFT rather than assumed — and it argues for treating the classifier's FM/AFM output as a screening prior, not a final answer, when selecting candidates for downstream simulation. If this first pass succeeds end-to-end for further candidates, the same pipeline extends to more of the shortlist (several of the higher-ranked ones are single-Fe-site and will need a supercell for AFM enumeration).

7. Summary of open items

Item	Status
MAGNDATA rescrape (2,414 entries)	Complete — 1,919 collected, 495 permanent failures (upstream data limits), 0 pending
MACE fine-tuning (60 epochs)	Complete — Stage 1 + Stage 2/SWA checkpoints saved, see §4.2
Ordering classifier retrain on real MAGNDATA structure labels	Complete — flat result, revealed a composition-only feature ceiling (§3)
Structure-aware classifier features (v2: symmetry/volume + local geometry)	Complete — accuracy 56.7% → 67.0%, balanced accuracy 60.7% → 68.1% (§3.1)
JARVIS ordering labels (currently 100% null, 19,116 records unusable for Phase 1)	Resolved as a byproduct of §4.1 — classifier now predicts ordering for all previously-unlabeled records, including JARVIS
Candidate screening pipeline (classifier → DMI filter → structure backfill → MACE relaxation)	Complete — 1,647 candidates pass the funnel (after the OQMD fix, §2), 298 relaxed (298/298 converged), top 30 in <code>outputs/phase4_shortlist.csv</code> (§4)
OQMD / 2DMPedia / NOMAD acquisition and merge	Complete — 3,178 records added (harmonized total 61,457 → 64,635); OQMD elements/space-group bug found and fixed (§2)
Real DFT magnetic-ordering validation (FM vs. AFM energy mapping)	Complete — mp-861559: DFT ground state is AFM, not FM ($\Delta E = -0.274$ eV/cell); classifier's FM call not confirmed, see §6.3
Phase 4 candidate selection (how many of the 30 shortlisted candidates to actually simulate)	Shortlist ready — first real DFT check (§6.3) argues for validating more candidates before committing to simulation
C2DB acquisition	Not started
Bond-dependent (Fert–Levy) DMI implementation for spin dynamics	Parked, needs a priority call
Simulated annealing via LAMMPS dynamics	Fixed and complete — root-caused to a LAMMPS <code>hybrid/overlay</code> bug (§6.1), workaround verified; refined 33-run finite-T scan complete, T_c signal inconclusive at current run length

Sections 2, 3, 4, and 5 (data harmonization, ordering classifier, candidate screening, MLIP fine-tuning) are now complete and their numbers final. Section 6 covers both the exploratory spin-dynamics run (§6.1–6.2, one hand-picked candidate system) and the real DFT magnetic-ordering validation campaign now underway (§6.3) — §4.4's shortlist is what both draw from.